

The Real Breadboard: Creating an Electronic Sandwich

A creative hands-on class project

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ABSTRACT

“The Real Breadboard: Creating an Electronic Sandwich” is a hands-on class that allows participants to experiment with electronic circuit techniques while exploring concepts surrounding electronic technology’s pervasive invasion into all aspects of public and private life.

Participants use electronic circuit prototyping techniques to create a satirical object that uses bread as a prototyping breadboard, exposing the absurdity accompanying certain technological “advancements” and our media-rich diets that creep into everything, including your sandwich. Participants will take away their creations and increase their appetite for conceptual/technical experimentation.

First developed in the isolation of an art studio, the expansion of this project into a participatory workshop conduces collaboration and broadens the possible interpretations of the inherent concepts while inviting the integration of new ideas from a larger and more diverse audience.

This hands-on class introduces participants to the project’s concept and prototyping using breadboards. The class provides electronic components such as microcontrollers and LEDs, from which participants build and code circuits to control an LED matrix. This circuit is built on actual bread to create the electronic sandwich with the components inside the sandwich as ingredients and the LEDs visible outside the sandwich through the bread.

Participants, provided with sample code and libraries, will write the code to make creative animations on the LEDs, which will appear on top of the sandwich. The result might be a pattern, a graphic drawing, or text. Participants combine their LED designs with laser-engraved images of their creation toasted into the bread, which also appears on the top slice of bread of the sandwich. The creative combination of these processes allows participants to explore exciting concepts. Inserting electronic consumerism into something meant for human internal consumption provides a thoughtful platform for participants to be creative and have fun.

In a world where a sandwich with grill marks that loosely resemble a religious icon sells online for a ludicrous amount, maybe it’s not so far-fetched to have electronic components in our food. Maybe your electronic sandwich’s LED matrix informs you of its

ingredients or nutritional content. Perhaps your electronic sandwich displays an attractive design, so you choose it over those boring sandwiches to its left and right that only show their price and expiration date. Possibly, there’s also a dark side. Is your electronic sandwich a data harvesting device relaying the contents of your bowels to advertisers and insurance agents? Or is it spewing misinformation and claiming a hotdog isn’t a sandwich?

As society continues to inundate every human experience and interaction with digital technology and media, we must reflect on these choices. Although this project is superfluous on its surface, the deeper goal is contemplation.

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